

What Is the Architecture and How It Works

Plain-English Perspective

The UM002B universal-memory chip — explained with desks, filing cabinets, and cameras. No engineering degree required.

A note before we begin

Every computer you have ever used has a split personality when it comes to memory: one part is fast but forgetful, the other is reliable but slow. The UM002B is an attempt to end that split — a single memory that is both.

This document explains what the chip is and how it works, using only everyday pictures. If you can picture an office desk, a filing cabinet, and a camera, you already have everything you need.

1. Why your computer feels slow: the two-memory shuffle

Picture your computer's memory as an office with exactly two pieces of furniture.

The desk is RAM. It's a big, clear surface, and anything sitting on it is instantly within reach. This is where all the real work happens — your open apps, your documents, your game, all spread out and ready. But the desk has one brutal rule: **when the lights go out, everything on it vanishes.** RAM needs constant power. Pull the plug, and the desk is wiped clean.

The filing cabinet is storage (your SSD). Everything filed inside is safe — lights on or off, it remembers. But nothing actually *happens* inside a filing cabinet. To use a file, someone has to walk over, find the right folder, and carry it back to the desk. And when you're done, it has to be carried back and re-filed, or the work is lost.

So your computer employs an invisible full-time mover, shuffling paper all day long:

- **Starting the computer** ("booting") is the mover carrying the entire working set from the cabinet to the desk, armload by armload.
- **Opening an app** is another trip to the cabinet.
- **Opening a file** is another trip.
- **Saving** is carrying the work back and re-filing it, just in case the lights go out.

Here is the part most people never realize: when your machine keeps you waiting — the spinning wheel, the loading bar, the slow wake from sleep — most of the time it isn't *thinking*. It's **moving furniture**. The processor itself is astonishingly fast; it spends much of its life standing around waiting for the mover to finish carrying things between the desk and the cabinet.

Every computer you have ever owned works this way. It is simply how computers were built.

2. The idea: a desk that can photograph itself

The UM002B starts from a simple question: *what if every spot on the desk had its own tiny camera — one that could photograph exactly what's there, need no power to keep the photograph, and put it all back when the lights return?*

That is what each UM002B memory cell is: **one cell, two halves.**

Half one: the capacitor — the desk surface. A capacitor stores a tiny amount of electric charge, and the pattern of that charge *is* the data. Reading the charge and rewriting it is blindingly fast — this is the half that does all of the daily work, everything you do on the computer. But charge is a restless thing. It leaks away slowly, like air seeping from a balloon, so the chip quietly tops it up over and over (a process called refresh). And if the power goes away entirely, the charge drains and the data goes with it. Fast — but forgetful. A perfect desk surface.

Half two: the memristor — the photographic memory. This half is built with a remarkable material called **hafnium oxide**. Instead of storing data as charge, it stores data in its **physical structure** — set into one arrangement or another, the way a light switch stays up or down. And here is the beautiful part: a light switch needs no electricity to stay where you left it. Flip it up, walk away for a year, come back — it's still up.

That sentence is the only physics in this entire document, and it's worth saying twice:

Charge leaks. Physical structure stays.

The memristor's job is not speed — it's *remembering without power*. It has exactly one job, and it only does it when the lights go out.

Two tiny switches (engineers call them transistors) sit in each cell alongside these halves. Think of them as doors: they decide, moment by moment, which half of the cell is connected and active. On a normal day, the doors open to the capacitor. When power fails, they swing the memristor into action.

So: every cell is both a lightning-fast desk surface *and* a camera that remembers without power. Thousands upon thousands of these cells, tiled into a vast array, make the chip.

3. A normal day: the camera sleeps

On an ordinary day, you will never meet the memristor.

You open apps, type documents, edit spreadsheets, play games — every bit of it runs on the capacitor side of each cell, at full speed. Reading and rewriting charge is the fastest thing the chip knows how to do, and it does nothing else all day.

Meanwhile, the memristor half of every cell simply **sleeps**, quietly holding the photograph that was taken during the last power cut. It is never consulted, never gets in the way, never slows anything down. It draws no attention to itself at all.

This division of labor is the heart of the architecture: **the fast half works, the remembering half waits**. The memristor doesn't make daily work faster — it doesn't need to. Daily work is already running at the capacitor's full speed. The memristor exists for a single moment — the moment the power fails — and it spends the rest of its life ready.

4. The lights go out: the photograph

Now the moment the whole design is built around.

The power fails — a blackout, a pulled plug, a drained battery. On an ordinary computer, this is the moment everything on the desk vanishes. On the UM002B, this is the moment the camera wakes up.

The chip has a **smart controller** built into it — think of it as a night watchman who never sleeps. The instant it senses the power failing, it acts, and it acts fast. It draws on a **small built-in energy reserve**: a little store of electricity kept on the chip for exactly one purpose, like the emergency lighting that clicks on in a hallway when the mains go out. It's only enough for one job — but one job is all that's needed.

That job: walk across the entire chip, cell by cell, and **translate each capacitor's charge pattern into the memristor's physical structure**. The fleeting is made lasting. The charge that would have leaked away is converted into an arrangement of matter that needs no power at all. Every desk surface, photographed.

When the photograph is done, the machine can go **fully, completely dark** — consuming nothing, remembering everything. The photographs don't need electricity. The physical structures just sit there, exactly as they were left, in total silence.

5. The lights come back: developing the photographs

Power returns. The night watchman reverses the process.

It reads each memristor's physical structure and **rebuilds the matching charge pattern on its capacitor** — developing every photograph back onto the desk, cell by cell, across the whole chip. Charge is restored where charge was; the desk is re-laid exactly as it had been.

And then the machine simply **continues** — exactly where it left off. The document is open at the same paragraph. The game is mid-level. The tabs are where they were. Nothing was carried anywhere, nothing was reloaded, nothing was rebuilt from a cabinet, because nothing ever left.

This is the experience the architecture is reaching for: **power loss stops being something you fear and becomes a pause button you never have to press.**

6. The bigger picture: one pool, no more shuffling

Step back, and the deepest change isn't the camera — it's what the camera makes possible.

Because every cell is both fast *and* remembering, **memory and storage stop being two different places.** There is no longer a desk *and* a cabinet. There is one pool of memory that does both jobs: it runs your programs at full speed, and it keeps your files without power.

And when there's only one pool, the full-time mover loses his job. Nothing is ever “loaded” from a cabinet to a desk, because there is no cabinet and no desk anymore — just one surface that is both.

Here's what daily life starts to feel like:

- **Starting the computer.** There is nothing to carry over from anywhere, so “booting” stops being a long caravan of loading and becomes simply waking up.
- **Opening apps and files.** They're already where they need to be. No fetching, no carrying, no loading bars while data is hauled into place.
- **Switching between things.** Everything you were doing is right there, because “right there” is the only place anything ever is.
- **One pool, shared.** The same memory that runs your programs also keeps your files. The boundary between “working memory” and “storage” — the boundary that created all that shuffling in the first place — quietly stops existing.

That is the whole of it. A cell with two halves: one that works at speed, one that remembers without power. A watchman that photographs the first onto the second when the lights fail, and develops it back when they return. And an entire computer, freed from a lifetime of carrying paper between a desk and a filing cabinet.

Glossary: the only six words you need

- **Cell** — the smallest unit of the chip. Each one holds a piece of data, with both a capacitor half and a memristor half inside.
 - **Capacitor** — the fast half of the cell. Stores data as electric charge. The desk surface.
 - **Memristor** — the remembering half of the cell. Stores data as a physical structure in hafnium oxide. The photograph. Needs no power to keep it.
 - **Hafnium oxide** — the special material inside the memristor that can be set into lasting physical arrangements.
 - **Controller** — the chip's built-in manager. Senses power failing and returning, and runs the photographing and developing. The night watchman.
 - **Energy reserve** — a small store of electricity kept on the chip, used for one thing only: taking the photograph when main power fails.
-

UM002B in one paragraph

The UM002B is a universal-memory chip: a single pool of memory that acts as both the computer's fast working space and its permanent keeper of files. Each of its cells pairs a tiny capacitor — extremely fast, doing all daily work — with a tiny hafnium-oxide memristor that remembers without power. A built-in controller watches the power: when it fails, it uses a small on-chip energy reserve to photograph every capacitor's state into its memristor as a lasting physical structure; when power returns, it develops those photographs back, and the machine picks up exactly where it left off. No separate RAM and storage, no shuffling between them — just one memory that is both.